

2021 Ph H2 Q2

Section: Our Dynamic Universe

Topic: Forces and the Doppler Effect

A train with steam engine and carriage accelerates. (a) Find tension in coupling. (b) Doppler effect: whistle frequency 511 Hz, observed 541 Hz, speed of sound 340 m/s. (i) Find speed of train. (ii) Comment on what frequency a passenger hears.

Worked solution

(a) Tension in coupling

Total mass = $9.75 \times 10^4 + 3.56 \times 10^4 = 1.331 \times 10^5$ kg
Acceleration $a = F/m = 1.15 \times 10^5 / 1.331 \times 10^5 = 0.864$ m s⁻²

Force needed to accelerate carriage only =
 $m_{\text{carriage}} \cdot a$

$$T = 3.56 \times 10^4 \times 0.864 = 3.08 \times 10^4 \text{ N}$$

Answer: 3.07×10^4 N

(b)(i) Speed of train

Formula: $f_{\text{obs}} = f_{\text{source}} \times v/(v - u)$
Rearr: $u = v(1 - f_{\text{source}}/f_{\text{obs}})$
 $= 340(1 - 511/541)$
 $= 18.9 \text{ m s}^{-1}$

Answer: 18.9 m s^{-1}

(b)(ii) The student is incorrect.

Passenger in carriage is moving with the train and whistle. No relative motion between source and passenger, so frequency heard = frequency emitted (511 Hz).

Final answers

(a) Tension $\approx 3.07 \times 10^4 \text{ N}$

(b)(i) Speed $\approx 18.9 \text{ m s}^{-1}$

(b)(ii) Incorrect — passenger hears 511 Hz (no Doppler shift).

Revision tips

- Newton's 2nd law: $F = ma$ for total acceleration.
- Coupling tension provides force only for carriage mass.
- Doppler effect: $f' = f v / (v - u)$ when source approaches observer.
- Relative motion matters: no relative motion $\rightarrow$ no shift.
- Always check whether observer or source is moving relative to medium.